

ANISH DHAR

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PROFESSIONAL SUMMARY

Early-career mechanical engineer with hands-on experience in pharmaceutical capital equipment design and competitive motorsport CAE. Took design ownership of an INR 4.5 Cr total-containment bin-washing system through to manufacturing release as part of a small team at Glatt Systems. Proficient in Autodesk Inventor & Vault PDM, GD&T, large assemblies (100+ parts), ANSYS FEA/thermal, and additive manufacturing. Exposure to ECN-driven change management, shop-floor manufacturing processes, and cross-functional project execution from concept through build.

EXPERIENCE

Engineering Trainee | Glatt Systems Pvt. Ltd.

Pune, IN · Jun 2025 – Present

- Took design ownership of a **total-containment bin-washing system** for potent API bins valued at **INR 4.5 Cr**; delivered complete design package through to manufacturing release, coordinating with procurement and the wider project team.
- Designed a novel **SS316 ducting joining system** for pharmaceutical process air supply; prototype tested at 2 bar in a single-connection qualification test, with the design targeting elimination of legacy flanged-joint assembly requirements.
- Contributed structural design to a **3-story all-steel vertical granulation line** for pre-production validation for a global pharmaceutical OEM; coordinated across mechanical, process, and controls sub-teams within a 4-member core team.
- Managed **engineering change notices (ECNs)** across active design packages; tracked part-level design changes through Vault PDM, coordinating with vendors and manufacturing on revision-controlled drawing releases.
- Gained direct **manufacturing floor exposure** across fabrication, assembly, and inspection stages; incorporated shop-floor feedback into design revisions and maintained GD&T drawing standards across assemblies of 100+ parts.

Engineering Intern – Thermals | Saarloha Advanced Materials Pvt. Ltd.

Pune, IN · Jun 2024 – Aug 2024

- Audited the water-cooling circuit of a **35T Electric Arc Furnace** and continuous casting line; quantified thermal losses from slag deposition on cooling panels via analysis of 30+ research papers and plant instrumentation data.
- Produced a structured improvement report with recommended design modifications to reduce parasitic heat losses in the EAF water-cooling loop.

ENGINEERING PROJECTS & COMPETITIONS

Thermal Systems Lead – e-BAJA & ATVC | VJTI Racing (SAE)

Mumbai, IN · Jun 2022 – Mar 2024

- Engineered a jet-impingement cooling enclosure for the motor controller using SLA/FDM additive manufacturing; **enclosure mass: 500 g**, targeting a **25°C reduction** in operating temperature versus baseline air-cooled configuration.
- Designed low-loss manifold geometry to minimise static pressure drop across a forced-air distribution network, reducing auxiliary battery load; achieved **AIR 1 Sales Presentation and AIR 5 Overall** at SAE e-BAJA 2024.
- Led low- and high-voltage system design for ISIE IKR 2023 EV entry; implemented multi-layer startup interlock sequence — awarded **Best Accelerating Vehicle, Best Business Proposal, AIR 5 EV category**.

Isopropyl Alcohol–Air Rocket Engine · Personal Project

- Designed a converging-diverging nozzle for **300 N sea-level thrust**; performed CNC machining of mild steel nozzle on a Jyoti VMC850, including fixturing design, toolpath programming, and post-test failure analysis.

CAPSTONE RESEARCH

Numerical Investigation of Lattice-Based Cooling Structures for High-Performance Thermal Management · Apr 2025

- Developed and meshed lattice unit-cell geometries in ANSYS; conducted mesh-convergence studies under compressive loading to characterise stress distribution and stiffness-to-weight behaviour in leading-edge cooling enhancement applications.

(Hons.) Vertically Graded Single-Unit Lattice Structures – Compressive Behaviour Modelling · May 2025

- Modelled graded-density lattice structures under uniaxial compression; extracted load-displacement and stress-strain data via ANSYS structural FEA to characterise energy absorption of graded versus uniform lattice architectures.

EDUCATION

B.Tech – Mechanical Engineering	VJTI Mumbai	CGPA: 7.67	Dec 2021 – May 2025
HSC – CBSE	City International School, Pune	88.2%	Jul 2021
SSC – ICSE	The Bishop's School, Pune	96.2%	May 2019

TECHNICAL SKILLS

CAD / PDM	Autodesk Inventor (large assemblies, sheet metal, weldments, GD&T; drawings), Vault PDM, Fusion 360, SolidWorks
Simulation	ANSYS Mechanical (structural, thermal; hex/tet meshing, mesh convergence), ANSYS Fluent (CFD), MATLAB/Simulink
Manufacturing	CNC Machining (Jyoti VMC850, Desk CNC), FDM & SLA additive manufacturing, mild steel & SS316 fabrication, fixturing, shop-floor exposure
Engineering	GD&T; (ASME Y14.5), tolerance stack-up, ECN/change management, FEA post-processing, technical documentation
Pharma Domain	Contained equipment design, potent API handling systems, pharmaceutical process air ducting, wash-cycle validation
Other	Python, MATLAB (CFD differencing schemes, 1D/2D heat transfer), topology optimisation, generative design